

Python CGI

1. Setup
 - a. Start Xampp->Apache
 - b. Check httpd.conf for the following
 - i. ScriptAlias /cgi-bin/ "D:/Xampp/cgi-bin/"
 - ii. <Directory "D:/Xampp/cgi-bin">
AllowOverride All
Options None
Require all granted
</Directory>
2. Verify Python path
 - a. python -c "import sys; print(sys.executable)"
3. create python file inside Xampp/cgi-bin folder and html file inside htdocs
 - a. example program1.py – GET Method
 - i.

```
#!C:\Users\Binu\AppData\Local\Programs\Python\Python311\python.exe
import cgi
print("Content-Type: text/html\n")
print("<html>")
print("<head><title>Python CGI Demo</title></head>")
print("<body>")
print("<h2><font color='red'>Hello from Python CGI</font></h2>")
# Handling form input
form = cgi.FieldStorage()
name = form.getvalue("name", "Guest")
print(f"<p>Welcome, {name}</p>")
print("</body>")
print("</html>")
```
 - ii. Run in browser <http://localhost/cgi-bin/program1.py?name=MCA>
 - b. Example program2.py & file1.html - POST method
 - i. File1.html inside htdocs folder
 1.

```
<html>
<head><title>POST Demo</title></head>
<body>
<form action="/cgi-bin/program2.py" method="post">
  <input type="text" name="name" id="name">
  <br><br>
  <input type="text" name="age" id="age">
  <br><br>
  <input type="submit" value="Submit">
</form>
</body>
</html>
```
 - ii. Program2.py inside cgi-bin folder
 1.

```
#!C:\Users\Binu\AppData\Local\Programs\Python\Python311\python.exe
import cgi
import cgiib
```

```

cgib.enable() # Enables error reporting
print("Content-Type: text/html\n")
form = cgi.FieldStorage()
name = form.getvalue("name", "Guest")
age = form.getvalue("age", "Unknown")
print("<html>")
print("<head><title>POST Method Demo</title></head>")
print("<body>")
print(f"<h2>Hello, {name}</h2>")
print(f"<p>Your age is {age}</p>")
print("</body>")
print("</html>")

```

- c. Run in browser <http://localhost/file1.html>

Lab Questions

Part A: Fundamentals

- Write a Python CGI script that prints "Hello, World!" in an HTML page.
 - Include the correct HTTP header.
 - Test it by accessing the script via your browser.
- Modify the script to display the current server time using Python's datetime module.
 - Output format: Current Time: HH:MM:SS.

Part B: GET Method

- Create a CGI script that accepts a query parameter name via the URL (GET method).
 - Example: http://localhost/cgi-bin/get_demo.py?name=Binumon
 - The script should greet the user by name.
- Extend the script to accept two parameters: num1 and num2.
 - Display their sum on the webpage.

Part C: POST Method

- Design an HTML form with fields for username and password.
 - Submit the form using POST to a CGI script.
 - The script should display the entered values.
- Enhance the script to validate input:
 - If username is empty, display an error message.
 - Otherwise, greet the user with their username.

Part D: Dynamic Content

- Create a CGI script that accepts multiple form fields (name, age, email) via POST.
 - Display the data in a formatted HTML table.
- Build a CGI script that generates a multiplication table for a number entered via POST.
 - Example: Input 5 → Output table from 5 × 1 to 5 × 10.

Part E: Error Handling & Integration

- Add error handling using the cgitb module.
 - Intentionally introduce a syntax error and observe how the error report is displayed in the browser.
- Create a feedback form (name, email, comments) that submits via POST.
 - The CGI script should save the feedback into a text file (feedback.txt) on the server.
 - Display a confirmation message after saving.